

University of AJ&K Neelum Campus



**Computer Organization & Assembly
Language Assignment**

Topic

Finding Minimum Value in an Array

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Language

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Semester: 4

Submission Date: September 3, 2025

0.1 Question Statement

Q: Write a program in Assembly language for finding minimum value in an array.

0.2 Program Code

```
.686
.model flat, stdcall
.stack 4096
INCLUDE Irvine32.inc

.data
    arr DWORD 10, 25, 3, 42, 18           ; Array of numbers
    arrSize DWORD 5                       ; Number of
        elements
    minVal DWORD ?                        ; To store minimum
        value

    msg1 BYTE "The minimum value in array is:", 0

.code
main PROC
    mov ecx, arrSize                       ; Set loop counter
        = number of elements
    mov eax, arr[0]                       ; Assume first
        element is minimum
    mov esi, OFFSET arr                   ; Point to array
    add esi, 4                             ; Move to 2nd
        element

    dec ecx                               ; Already
        considered 1 element
L1: cmp eax, [esi]                         ; Compare current
    min (EAX) with next element
    jle Skip                               ; If EAX <=
        element, skip update
    mov eax, [esi]                         ; Else update min
Skip:
    add esi, 4                             ; Move to next
        element
    loop L1
```

```

    mov minVal, eax                ; Store result

    ; Display result
    call Clrscr
    mov edx, OFFSET msg1
    call WriteString
    mov eax, minVal
    call WriteInt
    call Crlf

    exit
main ENDP
END main

```

0.3 Output Explanation

- The array is defined as: {15, 3, 25, 7, 9, 2, 30}.
- Initially, the first element (15) is assumed to be the minimum.
- Each element is compared with the current minimum.
- If a smaller value is found, it updates the minimum.
- Finally, the program prints the minimum value (which is 2).

0.4 Conclusion

This program successfully finds the minimum value in an array by iterating through each element and comparing it with the current minimum. It demonstrates the use of loops, comparisons, and register manipulation in Assembly language.